Multivariable Calculus: Problem Set 7

Detailed Assignment-Style Solutions with Diagrams

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How to read these solutions

Every problem is written in an assignment style: first the recognition cue, then the method, then the calculation, and finally the conclusion. I do not omit algebraic steps that are needed to justify the answer. For local extrema in two variables, the standard interior test is based on the Hessian matrix

$$H_f(x,y) = \begin{pmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{pmatrix}, \qquad D = f_{xx}f_{yy} - f_{xy}^2.$$

At an interior critical point, if $D > 0$ and $f_{xx} > 0$, the point is a local minimum; if $D > 0$ and $f_{xx} < 0$, the point is a local maximum; if $D < 0$, the point is a saddle point; if $D = 0$, the test is inconclusive.

$$\boxed{\textbf{Find } \nabla f = 0}$$

$$\Downarrow$$

$$\boxed{\textbf{Compute } D = f_{xx}f_{yy} - f_{xy}^2}$$

$$\swarrow \quad \Downarrow \quad \searrow$$

$$\boxed{D > 0,\ f_{xx} > 0\text{: local minimum}} \quad \boxed{D > 0,\ f_{xx} < 0\text{: local maximum}} \quad \boxed{D < 0\text{: saddle point}}$$

$$\Downarrow$$

$$\boxed{D = 0\text{: the Hessian test is inconclusive; use another method.}}$$

Contents

1 Problem 1: Positive definite matrices

Problem 1(i)

Let A be a 2×2 symmetric matrix. Show that A is positive definite if and only if all its diagonal entries and determinant are positive.

Method

Write a general symmetric 2×2 matrix as

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}.$$

The quadratic form is

$$q(x, y) = \begin{pmatrix} x & y \end{pmatrix} A \begin{pmatrix} x \\ y \end{pmatrix} = ax^2 + 2bxy + dy^2.$$

Positive definite means $q(x, y) > 0$ for every $(x, y) \neq (0, 0)$.

First assume that A is positive definite. Taking $(x, y) = (1, 0)$ gives

$$q(1, 0) = a > 0.$$

Taking $(x, y) = (0, 1)$ gives

$$q(0, 1) = d > 0.$$

So both diagonal entries are positive.

It remains to show that $\det A > 0$. Since $a > 0$, complete the square:

$$ax^2 + 2bxy + dy^2 = a\left(x + \frac{b}{a}y\right)^2 + \left(d - \frac{b^2}{a}\right)y^2.$$

Choose $x = -\frac{b}{a}y$ and choose $y \neq 0$. Since A is positive definite,

$$q\left(-\frac{b}{a}y, y\right) = \left(d - \frac{b^2}{a}\right)y^2 > 0.$$

Because $y^2 > 0$, this implies

$$d - \frac{b^2}{a} > 0.$$

Multiplying by $a > 0$ gives

$$ad - b^2 > 0.$$

Therefore

$$\det A = ad - b^2 > 0.$$

Conversely, assume that

$$a > 0, \qquad d > 0, \qquad ad - b^2 > 0.$$

We again complete the square:

$$q(x, y) = a\left(x + \frac{b}{a}y\right)^2 + \frac{ad - b^2}{a}y^2.$$

Here $a > 0$ and $\frac{ad-b^2}{a} > 0$. Hence both terms are nonnegative. If $q(x, y) = 0$, then both terms must be zero. Thus

$$y = 0, \qquad x + \frac{b}{a}y = 0.$$

Since $y = 0$, the second equation gives $x = 0$. Therefore $q(x, y) > 0$ for every nonzero vector (x, y). Thus A is positive definite.

Conclusion

For a symmetric matrix $A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}$,

$$A \text{ is positive definite} \quad \Longleftrightarrow \quad a > 0, \quad d > 0, \quad ad - b^2 > 0.$$

Problem 1(ii)

Let A be a 2×2 symmetric matrix. Show that A is negative definite if and only if all its diagonal entries are negative and determinant is positive.

Method

A matrix A is negative definite exactly when $-A$ is positive definite. So we apply the result from Problem 1(i) to $-A$.

Let

$$A = \begin{pmatrix} a & b \\ b & d \end{pmatrix}.$$

Then

$$-A = \begin{pmatrix} -a & -b \\ -b & -d \end{pmatrix}.$$

By Problem 1(i), $-A$ is positive definite if and only if

$$-a > 0, \qquad -d > 0, \qquad \det(-A) > 0.$$

The first two inequalities mean

$$a < 0, \qquad d < 0.$$

For a 2×2 matrix,

$$\det(-A) = (-a)(-d) - (-b)^2 = ad - b^2 = \det A.$$

Hence $\det(-A) > 0$ is the same as $\det A > 0$.

Conclusion

A symmetric 2×2 matrix is negative definite if and only if its diagonal entries are negative and its determinant is positive:

$$a < 0, \qquad d < 0, \qquad ad - b^2 > 0.$$

Problem 1(iii)

Let A be an $n \times n$ positive definite matrix. Show that all its diagonal entries are positive.

Method

Use the standard basis vector e_i, whose i-th coordinate is 1 and all other coordinates are 0.

Since A is positive definite,

$$x^T A x > 0$$

for every nonzero $x \in \mathbb{R}^n$. Choose

$$x = e_i = (0, \ldots, 0, 1, 0, \ldots, 0)^T.$$

Then $e_i \neq 0$, so

$$e_i^T A e_i > 0.$$

But $e_i^T A e_i$ selects exactly the (i, i) entry of A. Therefore

$$e_i^T A e_i = a_{ii}.$$

Thus

$$a_{ii} > 0$$

for every $i = 1, \ldots, n$.

Conclusion

Every diagonal entry of a positive definite matrix is positive:

$$a_{11} > 0, \ a_{22} > 0, \ldots, \ a_{nn} > 0.$$

2 Problem 2: Local extrema and saddle points

General method for Problem 2

For each differentiable function $f(x, y)$:

1. Compute the first partial derivatives f_x and f_y.
2. Solve the critical point equations $f_x = 0$ and $f_y = 0$.
3. Compute
$$D = f_{xx} f_{yy} - f_{xy}^2.$$
4. Classify each critical point using the Hessian test.
5. If a domain boundary is explicitly given, separate interior critical point classification from boundary behavior.

Problem 2(i)

$$f(x, y) = x^2 + xy + y^2 + y.$$

Compute the first partial derivatives:

$$f_x = 2x + y, \qquad f_y = x + 2y + 1.$$

Critical points satisfy

$$2x + y = 0, \qquad x + 2y + 1 = 0.$$

From the first equation,

$$y = -2x.$$

Substitute this into the second equation:

$$x + 2(-2x) + 1 = 0,$$

so

$$x - 4x + 1 = 0, \qquad -3x + 1 = 0, \qquad x = \frac{1}{3}.$$

Then

$$y = -2\left(\frac{1}{3}\right) = -\frac{2}{3}.$$

So the only critical point is

$$\left(\frac{1}{3}, -\frac{2}{3}\right).$$

Now compute second partial derivatives:

$$f_{xx} = 2, \qquad f_{yy} = 2, \qquad f_{xy} = 1.$$

Therefore

$$D = (2)(2) - 1^2 = 4 - 1 = 3 > 0.$$

Since $D > 0$ and $f_{xx} = 2 > 0$, the point is a local minimum.

The value is

$$f\left(\frac{1}{3}, -\frac{2}{3}\right) = \frac{1}{9} - \frac{2}{9} + \frac{4}{9} - \frac{2}{3}.$$

Put everything over 9:

$$\frac{1}{9} - \frac{2}{9} + \frac{4}{9} - \frac{6}{9} = \frac{1 - 2 + 4 - 6}{9} = -\frac{1}{3}.$$

Answer

The function has a local minimum at

$$\left(\frac{1}{3}, -\frac{2}{3}\right)$$

with value

$$-\frac{1}{3}.$$

There is no local maximum and no saddle point.

Problem 2(ii)

$$f(x, y) = x^3 + y^2 - 3x.$$

Compute the first partial derivatives:

$$f_x = 3x^2 - 3, \qquad f_y = 2y.$$

Set them equal to zero:

$$3x^2 - 3 = 0, \qquad 2y = 0.$$

Thus

$$x^2 = 1, \qquad y = 0.$$

So the critical points are

$$(1, 0), \qquad (-1, 0).$$

Second partial derivatives are

$$f_{xx} = 6x, \qquad f_{yy} = 2, \qquad f_{xy} = 0.$$

Hence

$$D = (6x)(2) - 0 = 12x.$$

At $(1, 0)$,

$$D = 12 > 0, \qquad f_{xx} = 6 > 0.$$

So $(1, 0)$ is a local minimum. Its value is

$$f(1, 0) = 1^3 + 0^2 - 3(1) = -2.$$

At $(-1, 0)$,

$$D = -12 < 0.$$

So $(-1, 0)$ is a saddle point. Its value is

$$f(-1, 0) = (-1)^3 + 0^2 - 3(-1) = -1 + 3 = 2.$$

Answer

Local minimum: $(1, 0)$ with value -2. Saddle point: $(-1, 0)$ with value 2. There is no local maximum.

Problem 2(iii)

$$f(x,y) = (x-y)(1-xy).$$

First expand the function because derivatives are easier after expansion:

$$f(x,y) = x - y - x^2y + xy^2.$$

Then

$$f_x = 1 - 2xy + y^2, \qquad f_y = -1 - x^2 + 2xy.$$

Critical points satisfy

$$y^2 - 2xy + 1 = 0, \qquad 2xy - x^2 - 1 = 0.$$

From the second equation,

$$2xy = x^2 + 1.$$

Substitute this into the first equation:

$$y^2 - (x^2+1) + 1 = 0.$$

Therefore

$$y^2 - x^2 = 0.$$

So

$$y = x \qquad \text{or} \qquad y = -x.$$

If $y = x$, then the equation $2xy = x^2 + 1$ becomes

$$2x^2 = x^2 + 1,$$

so

$$x^2 = 1.$$

Thus

$$(x,y) = (1,1), \qquad (-1,-1).$$

If $y = -x$, then

$$2x(-x) = x^2 + 1,$$

so

$$-2x^2 = x^2 + 1, \qquad -3x^2 = 1,$$

which has no real solution. Hence the only critical points are

$$(1,1), \qquad (-1,-1).$$

Now compute the Hessian entries:

$$f_{xx} = -2y, \qquad f_{yy} = 2x, \qquad f_{xy} = -2x + 2y.$$

At $(1,1)$,

$$f_{xx} = -2, \qquad f_{yy} = 2, \qquad f_{xy} = 0.$$

Thus

$$D = (-2)(2) - 0 = -4 < 0.$$

Therefore $(1,1)$ is a saddle point.

At $(-1,-1)$,

$$f_{xx} = 2, \qquad f_{yy} = -2, \qquad f_{xy} = 0.$$

Thus

$$D = (2)(-2) - 0 = -4 < 0.$$

Therefore $(-1,-1)$ is also a saddle point.

Answer

Both critical points are saddle points:

$$(1,1), \qquad (-1,-1).$$

There are no local maxima and no local minima.

Problem 2(iv)

$$f(x,y) = y^3 + 3x^2y - 6x^2 - 6y^2 + 2.$$

Compute first partial derivatives:

$$f_x = 6xy - 12x = 6x(y-2),$$

$$f_y = 3y^2 + 3x^2 - 12y = 3(x^2 + y^2 - 4y).$$

Critical points satisfy

$$6x(y-2) = 0, \qquad 3(x^2 + y^2 - 4y) = 0.$$

The first equation gives

$$x = 0 \qquad \text{or} \qquad y = 2.$$

The second equation gives

$$x^2 + y^2 - 4y = 0.$$

Case 1: $x = 0$. Then

$$y^2 - 4y = 0,$$

so

$$y(y-4) = 0.$$

Thus

$$y = 0 \quad \text{or} \quad y = 4.$$

This gives

$$(0,0), \qquad (0,4).$$

Case 2: $y = 2$. Then

$$x^2 + 2^2 - 4(2) = 0,$$

so

$$x^2 + 4 - 8 = 0, \qquad x^2 = 4.$$

Thus

$$x = 2 \quad \text{or} \quad x = -2.$$

This gives

$$(2,2), \qquad (-2,2).$$

Second partial derivatives are

$$f_{xx} = 6y - 12, \qquad f_{yy} = 6y - 12, \qquad f_{xy} = 6x.$$

At $(0,0)$,

$$f_{xx} = -12, \quad f_{yy} = -12, \quad f_{xy} = 0.$$

So

$$D = (-12)(-12) - 0 = 144 > 0, \qquad f_{xx} < 0.$$

Thus $(0,0)$ is a local maximum. Its value is

$$f(0,0) = 2.$$

At $(0,4)$,

$$f_{xx} = 12, \quad f_{yy} = 12, \quad f_{xy} = 0.$$

So

$$D = 144 > 0, \qquad f_{xx} > 0.$$

Thus $(0,4)$ is a local minimum. Its value is

$$f(0,4) = 4^3 - 6(4^2) + 2 = 64 - 96 + 2 = -30.$$

At $(2,2)$,

$$f_{xx} = 0, \quad f_{yy} = 0, \quad f_{xy} = 12.$$

Thus

$$D = 0 \cdot 0 - 12^2 = -144 < 0.$$

So $(2,2)$ is a saddle point.

At $(-2,2)$,

$$f_{xx} = 0, \quad f_{yy} = 0, \quad f_{xy} = -12.$$

Thus

$$D = 0 \cdot 0 - (-12)^2 = -144 < 0.$$

So $(-2,2)$ is a saddle point.

Answer

Local maximum: $(0,0)$ with value 2. Local minimum: $(0,4)$ with value -30. Saddle points: $(2,2)$ and $(-2,2)$.

Problem 2(v)

$$f(x,y) = xy(1-x-y).$$

Expand:

$$f(x,y) = xy - x^2y - xy^2.$$

Compute first partial derivatives:

$$f_x = y - 2xy - y^2 = y(1-2x-y),$$

$$f_y = x - x^2 - 2xy = x(1-x-2y).$$

Critical points satisfy

$$y(1-2x-y) = 0, \qquad x(1-x-2y) = 0.$$

We solve by cases.

If $x = 0$ and $y = 0$, we get $(0,0)$.

If $x = 0$ and $1 - 2x - y = 0$, then

$$1 - y = 0,$$

so $y = 1$, giving $(0,1)$.

If $y = 0$ and $1 - x - 2y = 0$, then

$$1 - x = 0,$$

so $x = 1$, giving $(1,0)$.

If

$$1 - 2x - y = 0, \qquad 1 - x - 2y = 0,$$

then

$$2x + y = 1, \qquad x + 2y = 1.$$

Subtract the second equation from the first:

$$x - y = 0,$$

so $x = y$. Substituting into $2x + y = 1$ gives

$$3x = 1, \qquad x = \frac{1}{3}.$$

Thus

$$y = \frac{1}{3}.$$

So the fourth critical point is

$$\left(\frac{1}{3}, \frac{1}{3}\right).$$

Second partial derivatives are

$$f_{xx} = -2y, \qquad f_{yy} = -2x, \qquad f_{xy} = 1 - 2x - 2y.$$

At $(0, 0)$,

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad D = -1 < 0.$$

So $(0, 0)$ is a saddle point.

At $(1, 0)$,

$$f_{xx} = 0, \quad f_{yy} = -2, \quad f_{xy} = -1.$$

Thus

$$D = 0(-2) - (-1)^2 = -1 < 0.$$

So $(1, 0)$ is a saddle point.

At $(0, 1)$,

$$f_{xx} = -2, \quad f_{yy} = 0, \quad f_{xy} = -1.$$

Thus

$$D = (-2)(0) - (-1)^2 = -1 < 0.$$

So $(0, 1)$ is a saddle point.

At $\left(\frac{1}{3}, \frac{1}{3}\right)$,

$$f_{xx} = -\frac{2}{3}, \quad f_{yy} = -\frac{2}{3}, \quad f_{xy} = 1 - \frac{2}{3} - \frac{2}{3} = -\frac{1}{3}.$$

Therefore

$$D = \left(-\frac{2}{3}\right)\left(-\frac{2}{3}\right) - \left(-\frac{1}{3}\right)^2 = \frac{4}{9} - \frac{1}{9} = \frac{1}{3} > 0.$$

Since $D > 0$ and $f_{xx} < 0$, this point is a local maximum.

Its value is

$$f\left(\frac{1}{3}, \frac{1}{3}\right) = \frac{1}{3} \cdot \frac{1}{3}\left(1 - \frac{1}{3} - \frac{1}{3}\right) = \frac{1}{9} \cdot \frac{1}{3} = \frac{1}{27}.$$

Answer

Local maximum: $\left(\frac{1}{3}, \frac{1}{3}\right)$ with value $\frac{1}{27}$. Saddle points: $(0, 0)$, $(1, 0)$, and $(0, 1)$. There is no local minimum.

Problem 2(vi)

$$f(x, y) = e^x \cos y.$$

Compute first partial derivatives:

$$f_x = e^x \cos y, \qquad f_y = -e^x \sin y.$$

A critical point would require

$$e^x \cos y = 0, \qquad -e^x \sin y = 0.$$

Since $e^x > 0$ for every real x, these equations reduce to

$$\cos y = 0, \qquad \sin y = 0.$$

But no real number y satisfies both $\sin y = 0$ and $\cos y = 0$ at the same time.

Answer

There are no critical points. Therefore the function has no local maxima, no local minima, and no saddle points in the usual interior critical-point sense.

Problem 2(vii)

$$f(x, y) = (x^2 + y^2)e^{y^2 - x^2}.$$

Let

$$r^2 = x^2 + y^2.$$

Then

$$f = r^2 e^{y^2 - x^2}.$$

Using the product rule,

$$f_x = 2xe^{y^2 - x^2} + r^2 e^{y^2 - x^2}(-2x).$$

Factor out $2xe^{y^2 - x^2}$:

$$f_x = 2xe^{y^2 - x^2}(1 - r^2).$$

Similarly,

$$f_y = 2ye^{y^2 - x^2} + r^2 e^{y^2 - x^2}(2y) = 2ye^{y^2 - x^2}(1 + r^2).$$

Critical points satisfy

$$2xe^{y^2 - x^2}(1 - r^2) = 0, \qquad 2ye^{y^2 - x^2}(1 + r^2) = 0.$$

Since $e^{y^2 - x^2} > 0$ and $1 + r^2 > 0$, the second equation gives

$$y = 0.$$

With $y = 0$, we have $r^2 = x^2$, and the first equation becomes

$$2xe^{-x^2}(1 - x^2) = 0.$$

Thus

$$x = 0 \qquad \text{or} \qquad x^2 = 1.$$

Therefore the critical points are

$$(0, 0), \qquad (1, 0), \qquad (-1, 0).$$

At $(0, 0)$, since

$$f(x, y) = (x^2 + y^2)e^{y^2 - x^2} \geq 0$$

and

$$f(0,0) = 0,$$

we immediately get a local minimum at $(0,0)$. In fact it is also a global minimum.

At $(1,0)$ and $(-1,0)$, we use the Hessian. From

$$f_x = 2xe^{y^2-x^2}(1-r^2), \qquad f_y = 2ye^{y^2-x^2}(1+r^2),$$

one obtains at $(\pm 1, 0)$:

$$f_{xx} = -\frac{4}{e}, \qquad f_{yy} = \frac{4}{e}, \qquad f_{xy} = 0.$$

Therefore

$$D = \left(-\frac{4}{e}\right)\left(\frac{4}{e}\right) - 0 = -\frac{16}{e^2} < 0.$$

So both $(1,0)$ and $(-1,0)$ are saddle points.

Answer

Local minimum: $(0,0)$ with value 0. Saddle points: $(1,0)$ and $(-1,0)$. There is no local maximum.

Problem 2(viii)

$$f(x,y) = e^y(y^2 - x^2).$$

Compute first partial derivatives:

$$f_x = e^y(-2x) = -2xe^y,$$

$$f_y = e^y(y^2 - x^2) + e^y(2y) = e^y(y^2 + 2y - x^2).$$

Critical points satisfy

$$-2xe^y = 0, \qquad e^y(y^2 + 2y - x^2) = 0.$$

Since $e^y > 0$, the first equation gives

$$x = 0.$$

Substitute $x = 0$ into the second equation:

$$y^2 + 2y = 0.$$

Thus

$$y(y+2) = 0,$$

so

$$y = 0 \qquad \text{or} \qquad y = -2.$$

The critical points are

$$(0,0), \qquad (0,-2).$$

Second partial derivatives are

$$f_{xx} = -2e^y, \qquad f_{xy} = -2xe^y,$$

$$f_{yy} = e^y(y^2 + 2y - x^2) + e^y(2y+2) = e^y(y^2 + 4y + 2 - x^2).$$

At $(0,0)$,

$$f_{xx} = -2, \quad f_{yy} = 2, \quad f_{xy} = 0.$$

Therefore

$$D = (-2)(2) - 0 = -4 < 0.$$

So $(0,0)$ is a saddle point.

At $(0,-2)$,

$$f_{xx} = -2e^{-2}, \quad f_{xy} = 0, \quad f_{yy} = e^{-2}(4-8+2) = -2e^{-2}.$$

Thus

$$D = (-2e^{-2})(-2e^{-2}) - 0 = 4e^{-4} > 0.$$

Since $f_{xx} < 0$, $(0,-2)$ is a local maximum. Its value is

$$f(0,-2) = e^{-2}((-2)^2 - 0) = \frac{4}{e^2}.$$

Answer

Local maximum: $(0,-2)$ with value $4/e^2$. Saddle point: $(0,0)$. There is no local minimum.

Problem 2(ix)

$$f(x,y) = y^2 - 2y\cos x, \qquad 1 \le x \le 7.$$

Complete the square in y:

$$f(x,y) = (y - \cos x)^2 - \cos^2 x.$$

This form is useful because for each fixed x, the smallest value in the y-direction occurs at $y = \cos x$.

For interior critical points, compute partial derivatives:

$$f_x = 2y\sin x, \qquad f_y = 2y - 2\cos x.$$

Set them equal to zero:

$$2y\sin x = 0, \qquad 2y - 2\cos x = 0.$$

The second equation gives

$$y = \cos x.$$

Substitute into the first equation:

$$2\cos x \sin x = 0.$$

Thus

$$\cos x = 0 \qquad \text{or} \qquad \sin x = 0.$$

For $1 < x < 7$, the relevant values are

$$x = \frac{\pi}{2}, \quad \pi, \quad \frac{3\pi}{2}, \quad 2\pi.$$

Hence the interior critical points are

$$\left(\frac{\pi}{2}, 0\right), \quad (\pi,-1), \quad \left(\frac{3\pi}{2}, 0\right), \quad (2\pi, 1).$$

Second partial derivatives are

$$f_{xx} = 2y\cos x, \qquad f_{xy} = 2\sin x, \qquad f_{yy} = 2.$$

At $(\pi,-1)$,

$$f_{xx} = 2(-1)(-1) = 2, \qquad f_{xy} = 0, \qquad f_{yy} = 2.$$

So

$$D = 2 \cdot 2 - 0 = 4 > 0, \qquad f_{xx} > 0.$$

Thus $(\pi,-1)$ is a local minimum, with value

$$f(\pi,-1) = (-1)^2 - 2(-1)\cos\pi = 1 + 2(-1) = -1.$$

At $(2\pi, 1)$,

$$f_{xx} = 2(1)(1) = 2, \qquad f_{xy} = 0, \qquad f_{yy} = 2.$$

So again it is a local minimum, with value

$$f(2\pi, 1) = 1 - 2(1)(1) = -1.$$

At $\left(\frac{\pi}{2}, 0\right)$,

$$f_{xx} = 0, \qquad f_{xy} = 2, \qquad f_{yy} = 2.$$

Thus

$$D = 0 \cdot 2 - 2^2 = -4 < 0.$$

So $\left(\frac{\pi}{2}, 0\right)$ is a saddle point.

At $\left(\frac{3\pi}{2}, 0\right)$,

$$f_{xx} = 0, \qquad f_{xy} = -2, \qquad f_{yy} = 2.$$

Thus

$$D = 0 \cdot 2 - (-2)^2 = -4 < 0.$$

So $\left(\frac{3\pi}{2}, 0\right)$ is a saddle point.

Boundary note

If local extrema are interpreted relative to the closed strip $[1, 7] \times \mathbb{R}$, then $x = 1$ is a boundary. The point $(1, \cos 1)$ is also a one-sided local minimum because $x \mapsto -\cos^2 x$ is increasing at $x = 1$. The right boundary $x = 7$ gives no local extremum at $(7, \cos 7)$ because moving left along the minimizing curve decreases the value while moving in the y-direction increases the value.

Interior answer

Interior local minima: $(\pi, -1)$ and $(2\pi, 1)$, both with value -1. Interior saddle points: $\left(\frac{\pi}{2}, 0\right)$ and $\left(\frac{3\pi}{2}, 0\right)$. There is no interior local maximum.

Problem 2(x)

$$f(x, y) = \sin x \sin y, \qquad -\pi \le x \le \pi, \qquad -\pi \le y \le \pi.$$

First classify the interior critical points. Compute

$$f_x = \cos x \sin y, \qquad f_y = \sin x \cos y.$$

Set them equal to zero:

$$\cos x \sin y = 0, \qquad \sin x \cos y = 0.$$

In the interior $(-\pi, \pi)^2$, the solutions are:

$$(0, 0),$$

and

$$\left(\frac{\pi}{2}, \frac{\pi}{2}\right), \quad \left(-\frac{\pi}{2}, -\frac{\pi}{2}\right), \quad \left(\frac{\pi}{2}, -\frac{\pi}{2}\right), \quad \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Second partial derivatives are

$$f_{xx} = -\sin x \sin y, \qquad f_{yy} = -\sin x \sin y, \qquad f_{xy} = \cos x \cos y.$$

At $(0, 0)$,

$$f_{xx} = 0, \quad f_{yy} = 0, \quad f_{xy} = 1,$$

so

$$D = 0 \cdot 0 - 1^2 = -1 < 0.$$

Thus $(0,0)$ is a saddle point.

At $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $\left(-\frac{\pi}{2}, -\frac{\pi}{2}\right)$,

$$\sin x \sin y = 1.$$

Therefore

$$f_{xx} = -1, \quad f_{yy} = -1, \quad f_{xy} = 0,$$

so

$$D = 1 > 0, \qquad f_{xx} < 0.$$

These are local maxima, both with value 1.

At $\left(\frac{\pi}{2}, -\frac{\pi}{2}\right)$ and $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

$$\sin x \sin y = -1.$$

Therefore

$$f_{xx} = 1, \quad f_{yy} = 1, \quad f_{xy} = 0,$$

so

$$D = 1 > 0, \qquad f_{xx} > 0.$$

These are local minima, both with value -1.

Boundary note for the closed square

On the closed square, weak one-sided boundary extrema also occur because $\sin(\pm\pi) = 0$. For example, on the edge $x = \pi$, points with $0 < y < \pi$ are weak local minima and points with $-\pi < y < 0$ are weak local maxima. Similar one-sided statements hold on the other edges. The main interior critical-point classification is the one above.

Interior answer

Local maxima: $\left(\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $\left(-\frac{\pi}{2}, -\frac{\pi}{2}\right)$, both with value 1. Local minima: $\left(\frac{\pi}{2}, -\frac{\pi}{2}\right)$ and $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, both with value -1. Saddle point: $(0,0)$.

3 Problem 3: Extreme values and least squares

Problem 3(i)

Find all local and global extreme values for

$$f(x, y) = xy(1 - x^2 - y^2)$$

on

$$[0, 1]^2.$$

Method

Because the domain is closed and bounded, a continuous function has global maximum and minimum values by the extreme value theorem. We check: interior critical points and boundary pieces.

Compute first partial derivatives:

$$f_x = y(1 - x^2 - y^2) + xy(-2x) = y(1 - 3x^2 - y^2),$$

$$f_y = x(1 - x^2 - y^2) + xy(-2y) = x(1 - x^2 - 3y^2).$$

In the interior $(0, 1)^2$, we have $x > 0$ and $y > 0$. Therefore the critical point equations reduce to

$$1 - 3x^2 - y^2 = 0, \qquad 1 - x^2 - 3y^2 = 0.$$

Equivalently,

$$3x^2 + y^2 = 1, \qquad x^2 + 3y^2 = 1.$$

Subtract the second equation from the first:

$$2x^2 - 2y^2 = 0.$$

Thus

$$x^2 = y^2.$$

Since $x, y > 0$ in the interior, this gives

$$x = y.$$

Substitute $y = x$ into $3x^2 + y^2 = 1$:

$$3x^2 + x^2 = 1, \qquad 4x^2 = 1, \qquad x = \frac{1}{2}.$$

Thus

$$y = \frac{1}{2}.$$

The only interior critical point is

$$\left(\frac{1}{2}, \frac{1}{2}\right).$$

Its value is

$$f\left(\frac{1}{2}, \frac{1}{2}\right) = \frac{1}{2} \cdot \frac{1}{2}\left(1 - \frac{1}{4} - \frac{1}{4}\right) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}.$$

Now examine the boundary.

On $x = 0$:

$$f(0, y) = 0.$$

On $y = 0$:

$$f(x, 0) = 0.$$

On $x = 1$:

$$f(1, y) = y(1 - 1 - y^2) = -y^3, \qquad 0 \le y \le 1.$$

So on this edge the minimum is -1 at $y = 1$, and the maximum is 0 at $y = 0$.

On $y = 1$:

$$f(x, 1) = x(1 - x^2 - 1) = -x^3, \qquad 0 \le x \le 1.$$

So on this edge the minimum is -1 at $x = 1$, and the maximum is 0 at $x = 0$.

Therefore the global minimum is

$$f(1, 1) = -1.$$

To prove that the interior value $\frac{1}{8}$ is the global maximum, observe that if $x^2 + y^2 > 1$, then

$$1 - x^2 - y^2 < 0,$$

so

$$f(x, y) \le 0.$$

Thus positive maxima can only occur where $x^2 + y^2 \le 1$. Let

$$s = x^2 + y^2.$$

By $2xy \le x^2 + y^2 = s$, we get

$$xy \le \frac{s}{2}.$$

Therefore

$$f(x, y) = xy(1 - s) \le \frac{s}{2}(1 - s).$$

For $0 \le s \le 1$,

$$\frac{s}{2}(1 - s)$$

is maximized at $s = \frac{1}{2}$, where its value is

$$\frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}.$$

Equality requires

$$x = y, \qquad x^2 + y^2 = \frac{1}{2},$$

which gives

$$x = y = \frac{1}{2}.$$

Region in $[0, 1]^2$	**Sign of** $1 - x^2 - y^2$	**Sign of** $f = xy(1 - x^2 - y^2)$
Inside the quarter disk $x^2 + y^2 < 1$	Positive	Positive, except on axes where $f = 0$
On the curve $x^2 + y^2 = 1$	Zero	Zero
Outside the quarter disk $x^2 + y^2 > 1$	Negative	Negative, since $x, y \ge 0$
Special points	$\left(\frac{1}{2}, \frac{1}{2}\right)$ gives max $\frac{1}{8}$	$(1, 1)$ gives min -1

Answer

Global maximum:

$$\frac{1}{8} \quad \text{at} \quad \left(\frac{1}{2}, \frac{1}{2}\right).$$

Global minimum:

$$-1 \quad \text{at} \quad (1, 1).$$

There are weak boundary local minima on the axes away from $(1, 0)$ and $(0, 1)$, because $f = 0$ there and nearby interior values in the quarter disk are nonnegative. The point $\left(\frac{1}{2}, \frac{1}{2}\right)$ is the only interior local maximum.

Problem 3(ii)

Let

$$f(x, y) = 3x^4 - 4x^2y + y^2.$$

Show that on every line $y = mx$, the function has a minimum at $(0, 0)$, but $(0, 0)$ is not a point of local minimum in $\mathbb{R}^2$.

Along the line

$$y = mx,$$

we substitute $y = mx$ into f:

$$f(x, mx) = 3x^4 - 4x^2(mx) + (mx)^2.$$

Therefore

$$f(x, mx) = 3x^4 - 4mx^3 + m^2x^2.$$

Factor out x^2:

$$f(x, mx) = x^2(3x^2 - 4mx + m^2).$$

At $x = 0$, this is 0. For fixed $m \neq 0$, the leading term near $x = 0$ is

$$m^2x^2,$$

which is positive for $x \neq 0$ sufficiently close to 0. Therefore $x = 0$ is a local minimum of the restricted one-variable function $x \mapsto f(x, mx)$.

If $m = 0$, then $y = 0$, and

$$f(x, 0) = 3x^4 \geq 0,$$

with equality at $x = 0$. Hence $x = 0$ is also a minimum along the line $y = 0$.

Now show that $(0, 0)$ is not a local minimum in the full plane. Complete the square:

$$f(x, y) = y^2 - 4x^2y + 3x^4.$$

Notice that

$$y^2 - 4x^2y + 4x^4 = (y - 2x^2)^2.$$

Thus

$$f(x, y) = (y - 2x^2)^2 - x^4.$$

Now approach $(0, 0)$ along the curve

$$y = 2x^2.$$

Then

$$f(x, 2x^2) = (2x^2 - 2x^2)^2 - x^4 = -x^4.$$

For every $x \neq 0$ close to 0,

$$-x^4 < 0 = f(0, 0).$$

So arbitrarily close to $(0, 0)$, the function is smaller than its value at $(0, 0)$.

Answer

On every straight line $y = mx$, the origin is a local minimum of the restricted one-variable function. But in the full plane, along the curve $y = 2x^2$, the values are $-x^4 < 0$. Hence $(0, 0)$ is not a local minimum in $\mathbb{R}^2$.

Problem 3(iii)

Given distinct real numbers $a_1, \ldots, a_n$ and real numbers $b_1, \ldots, b_n$, find the straight line that minimizes

$$f(x, y) = \sum_{i=1}^{n} (x + ya_i - b_i)^2.$$

Interpretation

The line has the form

$$b = x + ya.$$

Here x is the intercept and y is the slope. The expression $x + ya_i - b_i$ is the vertical error at the data point (a_i, b_i).

Define the following sums:

$$S_a = \sum_{i=1}^{n} a_i, \qquad S_b = \sum_{i=1}^{n} b_i, \qquad S_{aa} = \sum_{i=1}^{n} a_i^2, \qquad S_{ab} = \sum_{i=1}^{n} a_i b_i.$$

Compute the partial derivatives.

First,

$$f_x = 2 \sum_{i=1}^{n} (x + ya_i - b_i).$$

Set $f_x = 0$:

$$\sum_{i=1}^{n} (x + ya_i - b_i) = 0.$$

Therefore

$$nx + y \sum_{i=1}^{n} a_i - \sum_{i=1}^{n} b_i = 0,$$

so

$$nx + yS_a = S_b.$$

Second,

$$f_y = 2 \sum_{i=1}^{n} a_i (x + ya_i - b_i).$$

Set $f_y = 0$:

$$\sum_{i=1}^{n} a_i (x + ya_i - b_i) = 0.$$

Therefore

$$x \sum_{i=1}^{n} a_i + y \sum_{i=1}^{n} a_i^2 - \sum_{i=1}^{n} a_i b_i = 0,$$

so

$$xS_a + yS_{aa} = S_{ab}.$$

Thus (x, y) must solve the normal equations

$$\begin{pmatrix} n & S_a \\ S_a & S_{aa} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} S_b \\ S_{ab} \end{pmatrix}.$$

The determinant is

$$\Delta = nS_{aa} - S_a^2.$$

Since the a_i are not all equal, and in fact are distinct,

$$\Delta = n \sum_{i=1}^{n} (a_i - \bar{a})^2 > 0,$$

where

$$\bar{a} = \frac{1}{n} \sum_{i=1}^{n} a_i.$$

Therefore the solution is unique. By solving the 2×2 linear system,

$$x = \frac{S_b S_{aa} - S_a S_{ab}}{n S_{aa} - S_a^2},$$

$$y = \frac{n S_{ab} - S_a S_b}{n S_{aa} - S_a^2}.$$

Answer

The least-squares line is

$$b = x + ya,$$

where

$$x = \frac{S_b S_{aa} - S_a S_{ab}}{n S_{aa} - S_a^2}, \qquad y = \frac{n S_{ab} - S_a S_b}{n S_{aa} - S_a^2}.$$

4 Problem 4: Detailed study of $xy(1-x^2-y^2)$ on the square

Let

$$f(x,y) = xy(1-x^2-y^2)$$

on the square

$$[0,1]^2.$$

This is the same function as in Problem 3(i), but now the question asks for the classification more systematically.

Problem 4(i): Critical points

We already computed

$$f_x = y(1-3x^2-y^2), \qquad f_y = x(1-x^2-3y^2).$$

In the open square $(0,1)^2$, both x and y are positive, so the critical equations are

$$1-3x^2-y^2 = 0, \qquad 1-x^2-3y^2 = 0.$$

Solving these gives

$$x = y = \frac{1}{2}.$$

Therefore the only interior critical point is

$$\left(\frac{1}{2}, \frac{1}{2}\right).$$

If one solves $\nabla f = 0$ in the whole plane first, the full critical set in $\mathbb{R}^2$ is

$$(0,0),\ (1,0),\ (-1,0),\ (0,1),\ (0,-1),$$

$$\left(\frac{1}{2}, \frac{1}{2}\right), \left(\frac{1}{2}, -\frac{1}{2}\right), \left(-\frac{1}{2}, \frac{1}{2}\right), \left(-\frac{1}{2}, -\frac{1}{2}\right).$$

Among these, the points lying in $[0,1]^2$ are

$$(0,0), \quad (1,0), \quad (0,1), \quad \left(\frac{1}{2}, \frac{1}{2}\right).$$

But for the ordinary interior Hessian test on the square, only $\left(\frac{1}{2}, \frac{1}{2}\right)$ is an interior critical point.

Problem 4(ii): Local classification on $(0,1)^2$

Compute the second partial derivatives:

$$f_{xx} = -6xy, \qquad f_{yy} = -6xy, \qquad f_{xy} = 1-3x^2-3y^2.$$

At $\left(\frac{1}{2}, \frac{1}{2}\right)$,

$$f_{xx} = -6 \cdot \frac{1}{2} \cdot \frac{1}{2} = -\frac{3}{2},$$

$$f_{yy} = -\frac{3}{2},$$

$$f_{xy} = 1 - 3 \cdot \frac{1}{4} - 3 \cdot \frac{1}{4} = 1 - \frac{3}{4} - \frac{3}{4} = -\frac{1}{2}.$$

Therefore

$$D = \left(-\frac{3}{2}\right)\left(-\frac{3}{2}\right) - \left(-\frac{1}{2}\right)^2 = \frac{9}{4} - \frac{1}{4} = 2 > 0.$$

Since $f_{xx} < 0$, the point is a local maximum.

Answer on $(0,1)^2$

The only interior critical point is

$$\left(\frac{1}{2}, \frac{1}{2}\right),$$

and it is a local maximum with value

$$\frac{1}{8}.$$

There are no interior local minima and no interior saddle points.

Problem 4(iii): Local behavior on $[0,1]^2$

On the closed square, boundary points must be understood using relative neighborhoods inside the square.

The sign of f is controlled by

$$f(x,y) = xy(1 - x^2 - y^2).$$

Since $x \geq 0$ and $y \geq 0$ on the square, the sign depends on

$$1 - x^2 - y^2.$$

Thus:

$$f > 0 \quad \text{inside the quarter disk } x^2 + y^2 < 1,$$
$$f = 0 \quad \text{on the axes and on } x^2 + y^2 = 1,$$
$$f < 0 \quad \text{outside the quarter disk, within the square.}$$

Therefore:

1. The interior point $\left(\frac{1}{2}, \frac{1}{2}\right)$ is a local maximum.
2. Points on the axes $\{(x,0) : 0 \leq x < 1\}$ and $\{(0,y) : 0 \leq y < 1\}$ are weak local minima, because $f = 0$ there and nearby points sufficiently close to them inside the square have $f \geq 0$.
3. The point $(1,1)$ is a local minimum and also the global minimum.
4. Points on the quarter circle $x^2 + y^2 = 1$ with $x > 0, y > 0$ are not local extrema: nearby points inside the circle give $f > 0$, while nearby points outside the circle give $f < 0$.
5. The points $(1,0)$ and $(0,1)$ are also not local extrema because values of both signs occur arbitrarily close to them inside the square.

Answer on $[0,1]^2$

Local maximum: $\left(\frac{1}{2}, \frac{1}{2}\right)$. Weak boundary local minima: the axes segments $\{(x,0) : 0 \leq x < 1\}$ and $\{(0,y) : 0 \leq y < 1\}$, together with the corner $(1,1)$. Boundary sign-changing non-extreme points occur on the quarter circle $x^2 + y^2 = 1$ with $x, y > 0$, and at $(1,0)$ and $(0,1)$.

Problem 4(iv): Global maximum and minimum on $[0,1]^2$

From Problem 3(i), the global maximum is

$$\frac{1}{8}$$

at

$$\left(\frac{1}{2}, \frac{1}{2}\right).$$

The global minimum is

$$-1$$

at

$$(1, 1).$$

Final answer

$$\max_{[0,1]^2} f = \frac{1}{8} \quad \text{at} \quad \left(\frac{1}{2}, \frac{1}{2}\right),$$
$$\min_{[0,1]^2} f = -1 \quad \text{at} \quad (1, 1).$$

5 Problem 5: Extreme values with constraints

General method

When the constraint can be solved directly, reduce the problem to one variable. When the constraint is geometric, use Lagrange multipliers:

$$\nabla f = \lambda \nabla g.$$

Then also check whether the constraint set is compact. If the constraint is not compact, a maximum or minimum may fail to exist.

$$\boxed{\text{Constraint curve or surface: } g = 0}$$

At a constrained extremum, the level curve of f is tangent to the constraint.

Therefore the normal directions agree: ∇f is parallel to ∇g.

Hence the computational equation is $\boxed{\nabla f = \lambda \nabla g}$ together with $g = 0$.

Problem 5(i)

Find extreme values of

$$f(x, y) = xy$$

subject to

$$x + y = 1.$$

Use the constraint to write

$$y = 1 - x.$$

Then

$$f(x, 1 - x) = x(1 - x) = x - x^2.$$

Complete the square:

$$x - x^2 = -\left(x^2 - x\right) = -\left(\left(x - \frac{1}{2}\right)^2 - \frac{1}{4}\right) = \frac{1}{4} - \left(x - \frac{1}{2}\right)^2.$$

Thus

$$f(x, 1 - x) \leq \frac{1}{4},$$

with equality when

$$x = \frac{1}{2}.$$

Then

$$y = 1 - \frac{1}{2} = \frac{1}{2}.$$

As $x \to \infty$ or $x \to -\infty$, the expression $x - x^2$ tends to $-\infty$. Hence there is no minimum on the whole line.

Answer

Maximum value:

$$\frac{1}{4} \quad \text{at} \quad \left(\frac{1}{2}, \frac{1}{2}\right).$$

There is no minimum, because the line $x + y = 1$ is unbounded and $xy \to -\infty$ along it.

Problem 5(ii)

Find extreme values of

$$f(x, y) = x^2 + y^2$$

subject to

$$\frac{x}{a} + \frac{y}{b} = 1.$$

Assume $a \neq 0$ and $b \neq 0$.

Let

$$g(x, y) = \frac{x}{a} + \frac{y}{b} - 1.$$

Then

$$\nabla f = (2x, 2y), \qquad \nabla g = \left(\frac{1}{a}, \frac{1}{b}\right).$$

Lagrange multipliers give

$$(2x, 2y) = \lambda \left(\frac{1}{a}, \frac{1}{b}\right).$$

So

$$2x = \frac{\lambda}{a}, \qquad 2y = \frac{\lambda}{b}.$$

Therefore

$$x = \frac{\lambda}{2a}, \qquad y = \frac{\lambda}{2b}.$$

Substitute into the constraint:

$$\frac{1}{a} \cdot \frac{\lambda}{2a} + \frac{1}{b} \cdot \frac{\lambda}{2b} = 1.$$

Thus

$$\frac{\lambda}{2a^2} + \frac{\lambda}{2b^2} = 1.$$

Factor out $\lambda/2$:

$$\frac{\lambda}{2}\left(\frac{1}{a^2} + \frac{1}{b^2}\right) = 1.$$

Hence

$$\lambda = \frac{2}{\frac{1}{a^2} + \frac{1}{b^2}} = \frac{2a^2b^2}{a^2 + b^2}.$$

Therefore

$$x = \frac{1}{2a} \cdot \frac{2a^2b^2}{a^2 + b^2} = \frac{ab^2}{a^2 + b^2},$$
$$y = \frac{1}{2b} \cdot \frac{2a^2b^2}{a^2 + b^2} = \frac{a^2b}{a^2 + b^2}.$$

The minimum value is

$$x^2 + y^2 = \frac{a^2b^4}{(a^2 + b^2)^2} + \frac{a^4b^2}{(a^2 + b^2)^2}.$$

Factor:

$$x^2 + y^2 = \frac{a^2b^2(a^2 + b^2)}{(a^2 + b^2)^2} = \frac{a^2b^2}{a^2 + b^2}.$$

The constraint is a line, so it is unbounded. Along that line, $x^2 + y^2 \to \infty$. Hence there is no maximum.

Answer

Minimum value:

$$\frac{a^2b^2}{a^2 + b^2}$$

at

$$\left(\frac{ab^2}{a^2 + b^2}, \frac{a^2b}{a^2 + b^2}\right).$$

There is no maximum.

Problem 5(iii)

Find extreme values of

$$f(x, y) = x + y$$

subject to

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1.$$

Assume $a > 0$ and $b > 0$.

Let

$$g(x, y) = \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1.$$

Then

$$\nabla f = (1, 1), \qquad \nabla g = \left(\frac{2x}{a^2}, \frac{2y}{b^2}\right).$$

Lagrange multipliers give

$$(1, 1) = \lambda \left(\frac{2x}{a^2}, \frac{2y}{b^2}\right).$$

So

$$1 = \lambda \frac{2x}{a^2}, \qquad 1 = \lambda \frac{2y}{b^2}.$$

Therefore

$$x = \frac{a^2}{2\lambda}, \qquad y = \frac{b^2}{2\lambda}.$$

Substitute into the ellipse equation:

$$\frac{1}{a^2}\left(\frac{a^2}{2\lambda}\right)^2 + \frac{1}{b^2}\left(\frac{b^2}{2\lambda}\right)^2 = 1.$$

This becomes

$$\frac{a^2}{4\lambda^2} + \frac{b^2}{4\lambda^2} = 1.$$

So

$$\frac{a^2 + b^2}{4\lambda^2} = 1,$$

which gives

$$4\lambda^2 = a^2 + b^2.$$

Thus

$$2\lambda = \pm\sqrt{a^2 + b^2}.$$

If

$$2\lambda = \sqrt{a^2 + b^2},$$

then

$$x = \frac{a^2}{\sqrt{a^2 + b^2}}, \qquad y = \frac{b^2}{\sqrt{a^2 + b^2}}.$$

The value is

$$x + y = \frac{a^2 + b^2}{\sqrt{a^2 + b^2}} = \sqrt{a^2 + b^2}.$$

If

$$2\lambda = -\sqrt{a^2 + b^2},$$

then

$$x = -\frac{a^2}{\sqrt{a^2 + b^2}}, \qquad y = -\frac{b^2}{\sqrt{a^2 + b^2}}.$$

The value is

$$x + y = -\sqrt{a^2 + b^2}.$$

Answer

Maximum value:

$$\sqrt{a^2 + b^2}$$

at

$$\left(\frac{a^2}{\sqrt{a^2 + b^2}}, \frac{b^2}{\sqrt{a^2 + b^2}} \right).$$

Minimum value:

$$-\sqrt{a^2 + b^2}$$

at

$$\left(-\frac{a^2}{\sqrt{a^2 + b^2}}, -\frac{b^2}{\sqrt{a^2 + b^2}} \right).$$

6 Problem 6: Lagrange multipliers

General method

For an equality constraint $g = 0$, solve

$$\nabla f = \lambda \nabla g, \qquad g = 0.$$

After finding all candidates, evaluate f at each candidate. If the constraint set is compact, the largest and smallest candidate values are the global maximum and minimum. If the constraint set is not compact, separately check whether a maximum or minimum can fail to exist.

Problem 6(i)

$$f(x, y) = x^2 + y^2, \qquad g(x, y) = xy - 1 = 0.$$

The constraint is

$$xy = 1.$$

Compute gradients:

$$\nabla f = (2x, 2y), \qquad \nabla g = (y, x).$$

Lagrange equations are

$$2x = \lambda y, \qquad 2y = \lambda x, \qquad xy = 1.$$

Since $xy = 1$, neither x nor y is zero. Divide the first equation by the second equation:

$$\frac{2x}{2y} = \frac{\lambda y}{\lambda x}.$$

So

$$\frac{x}{y} = \frac{y}{x}.$$

Multiplying by xy gives

$$x^2 = y^2.$$

Thus

$$y = x \quad \text{or} \quad y = -x.$$

But $xy = 1 > 0$, so x and y must have the same sign. Therefore

$$y = x.$$

Substitute into $xy = 1$:

$$x^2 = 1.$$

Thus

$$x = 1 \quad \text{or} \quad x = -1.$$

So the candidates are

$$(1, 1), \qquad (-1, -1).$$

At both points,

$$f = x^2 + y^2 = 1 + 1 = 2.$$

Also, by the AM-GM inequality,

$$x^2 + y^2 \geq 2|xy| = 2$$

on the constraint. Hence 2 is the global minimum. There is no maximum because along $y = 1/x$,

$$f(x, 1/x) = x^2 + \frac{1}{x^2} \to \infty$$

as $x \to \infty$.

Answer

Minimum value 2 at $(1, 1)$ and $(-1, -1)$. No maximum exists.

Problem 6(ii)

$$f(x,y) = x^2 + 2y^2 - 4y, \qquad g(x,y) = x^2 + y^2 - 9 = 0.$$

The constraint is the circle

$$x^2 + y^2 = 9.$$

Compute gradients:

$$\nabla f = (2x, 4y - 4), \qquad \nabla g = (2x, 2y).$$

The Lagrange equations are

$$2x = \lambda(2x),$$
$$4y - 4 = \lambda(2y),$$
$$x^2 + y^2 = 9.$$

The first equation gives

$$(1 - \lambda)x = 0.$$

So either $x = 0$ or $\lambda = 1$.

Case 1: $x = 0$. Then the constraint gives

$$y^2 = 9,$$

so

$$y = 3 \quad \text{or} \quad y = -3.$$

The points are $(0, 3)$ and $(0, -3)$.

Case 2: $\lambda = 1$. The second equation becomes

$$4y - 4 = 2y.$$

Thus

$$2y = 4, \qquad y = 2.$$

The constraint gives

$$x^2 + 2^2 = 9, \qquad x^2 = 5.$$

Thus

$$x = \pm\sqrt{5}.$$

The points are

$$(\sqrt{5}, 2), \qquad (-\sqrt{5}, 2).$$

Evaluate f at all candidates:

$$f(0,3) = 0 + 2(9) - 12 = 6,$$
$$f(0,-3) = 0 + 2(9) + 12 = 30,$$
$$f(\pm\sqrt{5}, 2) = 5 + 2(4) - 8 = 5.$$

Since the circle is compact, the global minimum and maximum occur among these candidates.

Answer

Minimum value 5 at $(\sqrt{5}, 2)$ and $(-\sqrt{5}, 2)$. Maximum value 30 at $(0, -3)$. The point $(0, 3)$ is another constrained stationary point with value 6.

Problem 6(iii)

$$f(x,y) = y^2 - x^2, \qquad g(x,y) = \frac{1}{4}x^2 + y^2 - 1 = 0.$$

The constraint is

$$\frac{x^2}{4} + y^2 = 1.$$

A direct parametrization is easiest:

$$x = 2\cos\theta, \qquad y = \sin\theta.$$

Then

$$f(2\cos\theta, \sin\theta) = \sin^2\theta - (2\cos\theta)^2.$$

So

$$f = \sin^2\theta - 4\cos^2\theta.$$

Using $\sin^2\theta = 1 - \cos^2\theta$,

$$f = 1 - \cos^2\theta - 4\cos^2\theta = 1 - 5\cos^2\theta.$$

Since

$$0 \le \cos^2\theta \le 1,$$

we get

$$-4 \le f \le 1.$$

The maximum 1 occurs when

$$\cos^2\theta = 0,$$

so

$$\cos\theta = 0, \qquad x = 0, \qquad y = \pm 1.$$

The minimum -4 occurs when

$$\cos^2\theta = 1,$$

so

$$y = 0, \qquad x = \pm 2.$$

Answer

Maximum value 1 at $(0,1)$ and $(0,-1)$. Minimum value -4 at $(2,0)$ and $(-2,0)$.

Problem 6(iv)

$$f(x,y) = xy, \qquad g(x,y) = x^2 + 2y^2 - 6 = 0.$$

Compute gradients:

$$\nabla f = (y, x), \qquad \nabla g = (2x, 4y).$$

Lagrange equations are

$$y = 2\lambda x,$$

$$x = 4\lambda y,$$

$$x^2 + 2y^2 = 6.$$

If $x = 0$, then the second equation gives $0 = 4\lambda y$. On the constraint, $y \neq 0$, so this forces $\lambda = 0$. But then the first equation gives $y = 0$, contradiction. Hence $x \neq 0$. Similarly $y \neq 0$.

Substitute $y = 2\lambda x$ into $x = 4\lambda y$:

$$x = 4\lambda(2\lambda x) = 8\lambda^2 x.$$

Since $x \neq 0$,

$$1 = 8\lambda^2.$$

Thus

$$\lambda = \pm\frac{1}{2\sqrt{2}}.$$

From $y = 2\lambda x$,

$$y = \pm\frac{x}{\sqrt{2}}.$$

Use the constraint:

$$x^2 + 2\left(\frac{x^2}{2}\right) = 6.$$

Thus

$$2x^2 = 6, \qquad x^2 = 3.$$

So

$$x = \pm\sqrt{3}.$$

Then

$$y = \pm\sqrt{\frac{3}{2}},$$

with the sign depending on whether $y = x/\sqrt{2}$ or $y = -x/\sqrt{2}$.

If x and y have the same sign,

$$xy = \sqrt{3}\sqrt{\frac{3}{2}} = \frac{3}{\sqrt{2}}.$$

If x and y have opposite signs,

$$xy = -\frac{3}{\sqrt{2}}.$$

Answer

Maximum value:

$$\frac{3}{\sqrt{2}}$$

at

$$\left(\sqrt{3}, \sqrt{\frac{3}{2}}\right), \qquad \left(-\sqrt{3}, -\sqrt{\frac{3}{2}}\right).$$

Minimum value:

$$-\frac{3}{\sqrt{2}}$$

at

$$\left(\sqrt{3}, -\sqrt{\frac{3}{2}}\right), \qquad \left(-\sqrt{3}, \sqrt{\frac{3}{2}}\right).$$

Problem 6(v)

$$f(x, y) = x^2y^2, \qquad g(x, y) = x^2 + y^2 - 1 = 0.$$

Let

$$u = x^2, \qquad v = y^2.$$

Then

$$u \geq 0, \qquad v \geq 0, \qquad u + v = 1.$$

The objective becomes

$$f = x^2y^2 = uv.$$

Since $v = 1 - u$,

$$f = u(1-u) = \frac{1}{4} - \left(u - \frac{1}{2}\right)^2.$$

Thus the maximum is

$$\frac{1}{4}$$

when

$$u = \frac{1}{2}, \qquad v = \frac{1}{2}.$$

This means

$$x^2 = y^2 = \frac{1}{2},$$

so

$$x = \pm\frac{1}{\sqrt{2}}, \qquad y = \pm\frac{1}{\sqrt{2}}.$$

All four sign choices are allowed.

The minimum occurs when

$$uv = 0.$$

Thus either $u = 0$ or $v = 0$, meaning

$$x = 0 \quad \text{or} \quad y = 0.$$

Together with $x^2 + y^2 = 1$, the points are

$$(\pm 1, 0), \qquad (0, \pm 1).$$

Answer

Maximum value $\frac{1}{4}$ at

$$\left(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\right)$$

with all four sign combinations. Minimum value 0 at

$$(\pm 1, 0), \qquad (0, \pm 1).$$

Problem 6(vi)

$$f(x,y) = \sqrt{xy}, \qquad g(x,y) = x + y - c = 0.$$

Assume $c > 0$ and use the real-valued domain $xy \geq 0$. On the line $x + y = c$, the relevant nonnegative segment is

$$0 \leq x \leq c, \qquad y = c - x.$$

Then

$$f(x, c-x) = \sqrt{x(c-x)}.$$

Since square root is increasing, maximizing $\sqrt{x(c-x)}$ is the same as maximizing

$$x(c-x) = cx - x^2.$$

Complete the square:

$$cx - x^2 = \frac{c^2}{4} - \left(x - \frac{c}{2}\right)^2.$$

Therefore the maximum occurs at

$$x = \frac{c}{2}.$$

Then

$$y = c - \frac{c}{2} = \frac{c}{2}.$$

The maximum value is

$$f\left(\frac{c}{2}, \frac{c}{2}\right) = \sqrt{\frac{c}{2} \cdot \frac{c}{2}} = \frac{c}{2}.$$

The minimum on the closed nonnegative segment occurs at the endpoints:

$$(x, y) = (0, c) \quad \text{or} \quad (c, 0).$$

At both endpoints,

$$f = 0.$$

Answer

For $c > 0$, maximum value $c/2$ at $(c/2, c/2)$; minimum value 0 at $(0, c)$ and $(c, 0)$.

Problem 6(vii)

$$f(x, y) = x^2 + y^2, \qquad g(x, y) = x^4 + y^4 - 1 = 0.$$

Let

$$u = x^2, \qquad v = y^2.$$

Then

$$u \geq 0, \quad v \geq 0, \quad u^2 + v^2 = 1.$$

The objective is

$$f = u + v.$$

On the quarter circle $u^2 + v^2 = 1$, $u, v \geq 0$, the sum $u + v$ is minimized at the endpoints and maximized when $u = v$.

Minimum: endpoints

$$(u, v) = (1, 0) \quad \text{or} \quad (0, 1).$$

Then

$$f = u + v = 1.$$

The corresponding points are

$$(\pm 1, 0), \qquad (0, \pm 1).$$

Maximum: $u = v$. Since

$$u^2 + v^2 = 1, \qquad u = v,$$

we get

$$2u^2 = 1, \qquad u = \frac{1}{\sqrt{2}}.$$

Therefore

$$f = u + v = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \sqrt{2}.$$

Since

$$x^2 = u = \frac{1}{\sqrt{2}},$$

we get

$$x = \pm 2^{-1/4}.$$

Similarly

$$y = \pm 2^{-1/4}.$$

Answer

Minimum value 1 at $(\pm 1, 0)$ and $(0, \pm 1)$. Maximum value $\sqrt{2}$ at

$$(\pm 2^{-1/4}, \pm 2^{-1/4}),$$

with all four sign choices.

Problem 6(viii)

$$f(x, y) = x^4 + y^4, \qquad g(x, y) = x^2 + y^2 - 1 = 0.$$

Again let

$$u = x^2, \qquad v = y^2.$$

Then

$$u \geq 0, \quad v \geq 0, \quad u + v = 1,$$

and

$$f = u^2 + v^2.$$

Substitute $v = 1 - u$:

$$f = u^2 + (1 - u)^2.$$

Expand:

$$f = u^2 + 1 - 2u + u^2 = 2u^2 - 2u + 1.$$

Complete the square:

$$2u^2 - 2u + 1 = 2\left(u^2 - u\right) + 1 = 2\left(\left(u - \frac{1}{2}\right)^2 - \frac{1}{4}\right) + 1.$$

Thus

$$f = 2\left(u - \frac{1}{2}\right)^2 + \frac{1}{2}.$$

The minimum is

$$\frac{1}{2}$$

when

$$u = \frac{1}{2}.$$

Then

$$v = \frac{1}{2}.$$

So

$$x = \pm\frac{1}{\sqrt{2}}, \qquad y = \pm\frac{1}{\sqrt{2}}.$$

The maximum occurs at endpoints $u = 0$ or $u = 1$. Thus

$$f = 1.$$

The corresponding points are

$$(\pm 1, 0), \qquad (0, \pm 1).$$

Answer

Minimum value $\frac{1}{2}$ at

$$\left(\pm\frac{1}{\sqrt{2}}, \pm\frac{1}{\sqrt{2}}\right)$$

with all four sign choices. Maximum value 1 at $(\pm 1, 0)$ and $(0, \pm 1)$.

Problem 6(ix)

$$f(x_1, \ldots, x_n) = x_1 + x_2 + \cdots + x_n,$$

subject to

$$g(x_1, \ldots, x_n) = x_1^2 + x_2^2 + \cdots + x_n^2 - 1 = 0.$$

This is optimization of a linear function on the unit sphere.

Write

$$\mathbf{x} = (x_1, \ldots, x_n), \qquad \mathbf{1} = (1, \ldots, 1).$$

Then

$$f(\mathbf{x}) = \langle \mathbf{x}, \mathbf{1} \rangle .$$

The constraint is

$$\|\mathbf{x}\| = 1.$$

By the Cauchy-Schwarz inequality,

$$|\langle \mathbf{x}, \mathbf{1} \rangle| \leq \|\mathbf{x}\| \, \|\mathbf{1}\| .$$

Since

$$\|\mathbf{x}\| = 1, \qquad \|\mathbf{1}\| = \sqrt{n},$$

we get

$$|f(\mathbf{x})| \leq \sqrt{n}.$$

Therefore

$$-\sqrt{n} \leq f(\mathbf{x}) \leq \sqrt{n}.$$

Equality in Cauchy-Schwarz occurs when $\mathbf{x}$ is a scalar multiple of $\mathbf{1}$.

For the maximum, take

$$\mathbf{x} = \frac{1}{\sqrt{n}}(1, 1, \ldots, 1).$$

Then

$$f(\mathbf{x}) = n \cdot \frac{1}{\sqrt{n}} = \sqrt{n}.$$

For the minimum, take

$$\mathbf{x} = -\frac{1}{\sqrt{n}}(1, 1, \ldots, 1).$$

Then

$$f(\mathbf{x}) = -\sqrt{n}.$$

Answer

Maximum value:

$$\sqrt{n}$$

at

$$\left(\frac{1}{\sqrt{n}}, \ldots, \frac{1}{\sqrt{n}}\right).$$

Minimum value:

$$-\sqrt{n}$$

at

$$\left(-\frac{1}{\sqrt{n}}, \ldots, -\frac{1}{\sqrt{n}}\right).$$

Problem 6(x)

$$H(p_1, \ldots, p_n) = \sum_{i=1}^{n} p_i \log_2 p_i, \qquad p_i \geq 0,$$

subject to

$$p_1 + p_2 + \cdots + p_n = 1.$$

We use the convention

$$0 \log_2 0 = 0,$$

which is justified because

$$\lim_{t \to 0^+} t \log_2 t = 0.$$

For an interior critical point, assume $p_i > 0$ for all i. Let

$$g(p_1, \ldots, p_n) = p_1 + \cdots + p_n - 1.$$

The derivative of $p_i \log_2 p_i$ is

$$\frac{d}{dp_i}(p_i \log_2 p_i) = \log_2 p_i + \frac{1}{\ln 2}.$$

Lagrange multipliers give

$$\log_2 p_i + \frac{1}{\ln 2} = \lambda$$

for every i. Hence all p_i are equal:

$$p_1 = p_2 = \cdots = p_n.$$

Since they sum to 1,

$$p_i = \frac{1}{n}$$

for every i.

At the uniform point,

$$H\left(\frac{1}{n}, \ldots, \frac{1}{n}\right) = \sum_{i=1}^{n} \frac{1}{n} \log_2 \left(\frac{1}{n}\right).$$

There are n identical terms, so

$$H = n \cdot \frac{1}{n} \log_2 \left(\frac{1}{n}\right) = \log_2 \left(\frac{1}{n}\right) = -\log_2 n.$$

Because the function $t \mapsto t \log_2 t$ is convex on $[0, \infty)$, the sum H is convex on the probability simplex. Therefore the interior critical point is the global minimum.

The maximum of a convex function on the simplex occurs on the boundary, and here it is reached at a vertex. If one probability equals 1 and all others equal 0, then

$$H = 1 \cdot \log_2 1 + 0 = 0.$$

Since $p_i \log_2 p_i \leq 0$ for $0 \leq p_i \leq 1$, the value 0 is the largest possible value.

Answer

Minimum value:

$$-\log_2 n$$

at the uniform distribution

$$p_1 = p_2 = \cdots = p_n = \frac{1}{n}.$$

Maximum value:

$$0$$

at the vertices of the simplex, where one $p_j = 1$ and all other $p_i = 0$.

Final summary table

Problem	**Main result**
1	Positive definite and negative definite criteria for 2×2 symmetric matrices; positive definite matrices have positive diagonal entries.
2	Local extrema and saddle points obtained by solving $\nabla f = 0$ and applying the Hessian test.
3	$xy(1 - x^2 - y^2)$ has global maximum $1/8$ at $(1/2, 1/2)$ and global minimum -1 at $(1, 1)$ on $[0, 1]^2$; line-restricted minima need not imply a two-variable local minimum; least squares line is given by the normal equations.
4	Detailed classification of $xy(1 - x^2 - y^2)$ on the open square and closed square.
5	Constrained extrema solved by reducing constraints or using Lagrange multipliers.
6	Lagrange multiplier candidates evaluated on each constraint; compactness determines whether global extrema must exist.